

Problem Set 5

ECON 2010 – Macroeconomics

Spring 2019

Due at the start of class Week 13, Wednesday, April 17. You have **two weeks** to complete this problem set. No late problem sets will be accepted. Please refer to the syllabus section on late assignments.

This problem set will work the same as the first problem set and all problem sets. Work on a separate sheet of paper on your own, and we will correct (some of) the problem set together.

Name:

Problem set instructions.

1. **Do not write on this problem set sheet** until we go over the problem set during class. You will use this problem set sheet to correct the answers that you provide on the separate sheet when we go over the problem set in class the day that it is due.
 2. Write your final answers to each question on a separate sheet of paper.
 3. Write your final answers in pen.
 4. Ensure that your name appears on this problem set and on the separate sheet(s) of paper with your answers, and staple your answers to the back of this problem set.
 5. When appropriate, show your work for each problem. Even if your answer is technically incorrect, you may get many points for the work you did to arrive at that answer. If in doubt, show your work.
 6. Note that not all questions have *one* right answer. Some questions might have many correct answers.
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Problem 1 (12 points)

Follow the instructions for each of the following statements. Hold all else fixed when you consider the effect of any changes described in the statement. Correctly identifying whether the statement is true or false is worth 1 point. The explanation is worth 3 points.

To remind you, an explanation needs to give *reasons* why. An explanation is not an explanation if it simply restates the original statement in slightly different words. For example, the reason that the Earth orbits the sun is *not* because the Earth goes around the sun! The *reason* that the Earth orbits the sun is that the sun is more massive than the Earth, it warps space time with its mass, and pulls the Earth towards it using gravitational force. The Earth orbits, rather than falling into the sun because the Earth has countervailing perpendicular velocity that it acquired at the dawn of the solar system. Er, anyway...

An additional reminder: When you explain your answer, *draw on material that you've learned in class thus far*. (As is almost *always* the case) You don't need any information outside of what we've learned in class thus far to offer a correct explanation, and in fact information from outside of class is extremely unlikely to get you to the correct answer and explanation.

- (a) Indicate whether the following statement is **true or false**, and in just a few sentences, **briefly explain why**. (4 points)

If I own a Picasso, and the value of that Picasso increases, my income has gone up.

- (b) Indicate whether the following statement is **true or false**, and in just a few sentences, **briefly explain why**. (4 points)

When an economy operates at full employment, its unemployment rate is zero.

- (c) Indicate whether the following statement is **true or false**, and in just a few sentences, **briefly explain why**. (4 points)

*One of the **reasons** that the aggregate demand (AD) curve is downward sloping is because when the aggregate price level goes up, the real value wages goes down, and so individuals have less income to consume, and they demand fewer (total) goods and services (output).*

Problem 2 (16 points)

In this problem, you'll explain basic features of the aggregate supply and aggregate demand curves, and the way money moves through the economy.

- (a) **In your own words**, explain why the long run aggregate supply (LRAS) “curve” is a vertical line. For full credit, your answer must mention *price levels* (or *the price level*), *input prices*, *output* and *producers*. (3 points)

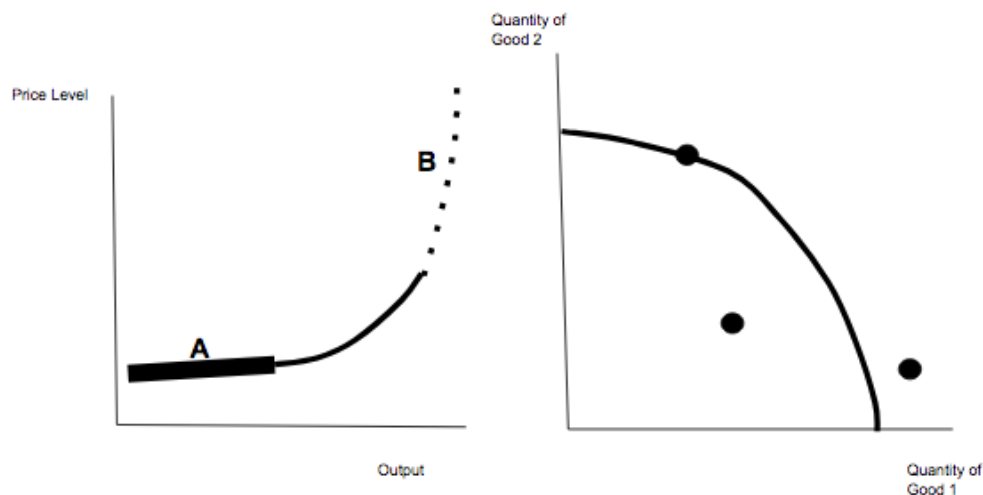
- (b) **In your own words**, explain why the short run aggregate supply (SRAS) is upward sloping. For full credit, your answer must mention *price levels* (or *the price level*), *input prices*, *output* and *producers*. (3 points)

- (c) Consider the following plot of the short run aggregate supply (SRAS) curve. Two parts of the curve are differentiated and labeled for you already. The first is Part A (bold); the second is Part B (dashed). In addition to the SRAS curve, you are also given a production possibilities frontier (PPF) curve with three unlabeled points.

Label the points on the PPF plot that conceptually correspond to A and B on the SRAS plot. Then, **explain briefly** why those points are conceptually related to what is happening in Part A and Part B of the SRAS curve. Note that there are three points but only two letters. You may label *all* of the points (that is, use one letter twice), or you may choose to *only* label two of the three points. Do not label a single point with both A and B.

Finally, note that the PPF is drawn for only two goods (or services), while the SRAS curve

represents the entire macroeconomy (many goods and services). The plots are still related — the PPF has just been simplified (use of a *simplifying assumption*) so that we can actually visualize the relationship and work with the curve. So, even though the number of goods/services represented by each plot are not the same, the two plots in fact are still *conceptually* closely related. **(5 points)**



- (d) Consider the following. The government taxes 20% of earnings. Individuals have a marginal propensity to consume of 0.7 after taxes. And individuals spend 10% of their after-tax income on imports. If the population of our economy is just 10 people, each of whom earn \$10,000 in a given year, by how much will the total *consumption* component of GDP increase by in one year. Hold all else fixed when you consider this problem, and do *not* consider any multiplier effects. **(5 points)**

Problem 3 (33 points)

Background Information:

Note that this background is about setting the stage for the problem; you do *not* need to use this information to answer the questions in this problem.

The Financial Crisis of 2007 (if you don't recall) was a financial crisis that developed first in the home mortgage market (i.e. the home loan market), and eventually spread throughout financial markets in the US (and eventually the globe). The crisis further precipitated the Great Recession of 2008. There isn't enough room here to go into a detailed timeline of what happened and why, but it's worth setting the stage for the problem with a little bit of information. I happen to think it makes the problem more interesting if you think about the history surrounding it.

The Financial Crisis was preceded by a long period of low interest rates in the US such that banks and financial institutions had to try to find new people to lend money to. This was done primarily by finding new people to provide home loans to. Note that we have not (yet) talked about interest rates in this class, and in case you aren't familiar, an interest rate is just a price to borrow money. It's usually expressed as a small fraction that represents what you have to pay in addition to the money you borrow. When interest rates are low, it's relatively inexpensive to borrow money — that is, you don't have to pay as much to borrow money. Whereas when interest rates are high, it's relatively expensive to borrow money.

Many of the new home loans that were made during this period of low interest rates were made to people who ordinarily wouldn't have qualified for a home loan. For example, these were people who had low incomes (and so would have a hard time paying back the loan), or bad credit (a history of not paying back loans). These loans to low income or bad credit borrowers were referred to as *subprime* loans and were especially risky. By risky, I mean that there was a relatively high probability that they weren't paid going to be paid back.

Usually, this riskiness translates directly into higher prices to borrow money — that is, more risky borrowers pay higher interest rates. And in some respects it did — many of these subprime loans were structured with low initial interest rates, but rates that jumped up *very* high after a few months or years to reflect the risk of the loan. (Loans with these low “teaser” interest rates are frequently cited as an example of predatory lending practices by banks. You may have heard them called *adjustable* rate mortgages.) Individual banks made these loans, but did not typically hold onto them over the long run (life of the loan). Instead of collecting mortgage payments themselves, they sold their loans to other financial institutions, oftentimes large investment banks, that in turn, bundled together lots of loans and sold the rights to collect future interest payments on those bundled loans once again. For reasons that are still not fully understood, banks frequently did not properly account for the riskiness of the loans that they were trading.

This system of risky lending and borrowing worked for awhile because housing prices kept increasing. Even if a home buyer could no longer afford their monthly mortgage payments, they could sell their house at a profit (relative to what they bought it at), repay their entire mortgage and incur no lasting debt. But home prices did not keep rising indefinitely — they eventually fell, and

when they did so, the consequences rippled through the economy.

Homeowners began defaulting on their loans, causing banks to repossess their homes because of non-payment. And even when they didn't default, they suffered huge losses to their wealth as they struggled to make large payments on an asset (their house) that was worth less and less relative to what they had paid for it. Banks and other financial institutions that had bought financial products that paid them streams of future mortgage interest payments were suddenly not being paid because homeowners weren't making their mortgage payments. Many banks had purchased enormous quantities of these financial products (called, for example, subprime mortgage interest bonds, collateralized debt obligations, and credit default swaps), had not properly assessed their risk, and when they became worthless, the banks struggled to cover their expenses, and lacked capital to make new loans to those who still wanted them. The financial crisis is oftentimes called a "credit crunch" because banks lacked the capital to make new loans, which causes economic activity to stall.

Some banks that had invested heavily in these financial products backed by risky home mortgages actually collapsed. Lehman Brothers, an investment bank in New York that collapsed in 2007 and eventually declared bankruptcy in 2008, is a prominent example. Many smaller banks and institutions that originated many of the risky loans themselves such as Countrywide and Indy-Mac also declared bankruptcy. But it was the larger banks and financial institutions — Goldman Sachs, Merrill Lynch, Morgan Stanley, etc. — that were deemed by the government "too big to fail." If even one of those large banks collapsed, the government reasoned, the loss of credit could plunge the economy into a full blown depression, not merely a recession. The Bush and then Obama administrations used a variety of techniques in order to ensure that those "too big to fail" institutions did not go bankrupt and continued to keep financial markets working; that is, continued to offer loans.

OK, now on to the actual problem...

The Problem:

Consider aggregate supply and aggregate demand in the US at the time of the Financial Crisis of 2007 and the Great Recession of 2008, and answer the following questions.

- (a) According the standard definition of economists, what makes the Great Recession a *recession*?
(1 point)

(b) One prominent explanation for the Great Recession is that during the financial crisis there was a substantial reduction in housing prices that increased consumer debt (because many people owed more on their houses than their house was worth) and reduced consumer *wealth* in the US. That is, at a given price level, the Great Recession caused a reduction in consumer wealth because of a drop in housing prices. Hold all else fixed, and consider how this reduction in consumer wealth brought about by the housing market collapse affected the macroeconomy. **(16 points total)**

(1) Which curve shifts as a result of this change? You can say **long run aggregate supply**, **short run aggregate supply**, **aggregate demand**, **none**, or you may choose *multiple curves*. **(2 points)**

(2) Which direction do the curve(s) shift? For **each** curve you said would shift above, you can say **right** or **left**. If no curves shift, just write **none**. **(2 points)**

(3) In just a few sentences, **briefly explain why** the curve(s) will shift. If no curves shift, just write **none**. **(5 points)**

- (4) **Draw** a plot with long run aggregate supply, short run aggregate supply, and aggregate demand in which the economy is in *long run equilibrium*. Then, use dashed lines to **draw** new curve(s) that represent any curves that shift as a result of the change described above. **Label your axes and your curves. Label** the starting equilibrium with an **A**, and the new *short run* equilibrium that results from any curve shifts with a **B**. If no curves shift, just draw the three curves in equilibrium and write “no curves shift”. **(5 points)**

- (5) Indicate how output *and* the price level have changed from the starting equilibrium A to the new short run equilibrium B. If no curves shifted and the equilibrium did not change, just write none. **(2 points)**

- (c) Another prominent explanation for the Great Recession is that the Financial Crisis involved a stock market crash. Between 2007 and 2009, the Dow Jones Industrial Average (an often-used measure of the value of stocks in the US) dropped from a value of 14,000 to 6,600. **Consider this drop in the stock market as a change in consumer wealth and consumer confidence.** That is, at a given price level, the Great Recession caused a reduction in consumer wealth because of a decline in the stock market. Hold all else fixed, and consider how this reduction in the value of the stock market affected the macroeconomy. **(16 points total)**

Note: To answer this question, you might need to understand what the stock market *is*, and I recognize that not everyone does. So, here is a brief explanation that I think is sufficient to allow you to consider how this stock market crash from 2007-2009 affected the macroeconomy. The stock market is a market (yes! like markets for goods and services that we’ve talked about before in class! exactly like those!) in which there are sellers and buyers of a product.

The product in this market is a small ownership *share* in a company. All ownership shares together are called the company's *stock*.

Companies issue stock (typically) once during what is called an *initial public offering* (IPO). Companies conduct an IPO to raise money — they trade ownership in their company for cash at the time of the IPO. Regular individuals like you or me (or institutions that represent a group of regular individuals) buy the stocks at the IPO. (Well, in theory it's regular individuals, and you should think it's regular individuals for this problem, but you actually have to have the right connections to buy in an IPO.) The IPO is the only point at which the company makes money *directly* from their stock.

After the initial shares are sold in the IPO, the shares are bought and sold by *anyone* in the stock market. There is a demand curve for shares in a given stock, a supply curve, and at any moment in time there is an equilibrium at which there is a share price and a share quantity sold. Remember, just like there is an equilibrium in any market! Those prices are what you see listed on a stock exchange like the New York Stock Exchange. (Things like the Dow Jones Industrial Average is an average of a bunch of different stock prices.) Note that while a shares in a stock have prices, those prices do not reflect a final good that is produced that year; thus stocks are not part of the average price level.

Regular people (or institutions that represent regular people like retirement savings plans) own shares of companies in the stock market. I can own shares; you can own shares; your grandmother can own shares. If we own shares we can sell our shares. And if you don't own shares, you can buy shares. So, bottom line for this problem — the people who own shares and who are affected when the stock market does better or worse are, in effect, regular people like you and me.

- (1) Which curve shifts as a result of this change? You can say **long run aggregate supply**, **short run aggregate supply**, **aggregate demand**, **none**, or you may choose *multiple curves*. **(2 points)**
- (2) Which direction do the curve(s) shift? For **each** curve you said would shift above, you can say **right** or **left**. If no curves shift, just write **none**. **(2 points)**

- (3) In just a few sentences, **briefly explain why** the curve(s) will shift. If no curves shift, just write **none**. **(5 points)**
- (4) **Draw** a plot with long run aggregate supply, short run aggregate supply, and aggregate demand in which the economy is in *long run equilibrium*. Then, use dashed lines to **draw** new curve(s) that represent any curves that shift as a result of the change described above. **Label** your axes and your curves. **Label** the starting equilibrium with an **A**, and the new *short run* equilibrium that results from any curve shifts with a **B**. If no curves shift, just draw the three curves in equilibrium. **(5 points)**
- (5) Indicate how output *and* the price level have changed from the starting equilibrium A to the new equilibrium B. If no curves shifted and the equilibrium did not change, just write none. **(2 points)**

Problem 4 (20 points)

In the previous problem, you considered the consequences of the financial crisis for the macroeconomy. Here you'll consider how the government response to the crisis and the Great Recession was designed to work. The primary *fiscal policy response* was an economic stimulus package in 2009 that involved the following: tax cuts for individuals, infrastructure projects, government transfers to state governments to pay for health and other services. The stimulus package cost \$840 billion. For reference, US GDP in 2009 was \$14.4 trillion, and the overall size of the US government budget in 2009 (in terms of expenditure) was \$3.1 trillion.

Evaluating the effect of the stimulus package is exceedingly difficult to do (page 2019 of your textbook describes a little bit about why). But in this problem we can consider why it was designed the way that it was, what it sought to accomplish, and how it might have accomplished those objectives.

- (a) Fiscal policy refers to the combination of which two things that governments can do to affect the macroeconomy? **(1 point)**
- (b) Consider the component of the stimulus package that came in the form of increased infrastructure spending by the federal government. Here is one example of that infrastructure spending from a NYTimes article on the topic: “The nuclear cleanup at the Savannah River Site in Aiken, S.C.... received \$1.6 billion in stimulus money. As soon as the money arrived in the summer of 2009, the retired cold war nuclear plant hired thousands of workers to decommission reactors, install pumps in the liquid waste tanks and ship barrels of solid waste to a salt formation in the Chihuahuan Desert. Workers... filled up nearly all of the area’s apartments, hotels and restaurants.” Infrastructure projects like this occurred around the country. In the problem, consider the totality of government infrastructure spending, not just this one example. **(16 points total)**

- (1) Which curve shifts as a result of increased infrastructure spending by the federal government? You can say **long run aggregate supply**, **short run aggregate supply**, **aggregate demand**, **none**, or you may choose *multiple curves*. (2 points)
- (2) Which direction do the curve(s) shift? For **each** curve you said would shift above, you can say **right** or **left**. If no curves shift, just write **none**. (2 points)
- (3) In just a few sentences, **briefly explain why** the curve(s) will shift. If no curves shift, just write **none**. (5 points)
- (4) **Draw** a plot with long run aggregate supply, short run aggregate supply, and aggregate demand in which the economy is in *long run equilibrium*. Then, use dashed lines to **draw**

new curve(s) that represent any curves that shift as a result of the change described above. **Label** your axes and your curves. **Label** the starting equilibrium with an **A**, and the new *short run* equilibrium that results from any curve shifts with a **B**. If no curves shift, just draw the three curves in equilibrium. **(5 points)**

- (5) Indicate how output *and* the price level have changed from the starting equilibrium A to the new equilibrium B. If no curves shifted and the equilibrium did not change, just write none. **(2 points)**

- (c) If the marginal propensity to consume in the U.S is 0.4, **calculate the fiscal multiplier. (1 point)**

- (d) If the fiscal multiplier you calculated above worked perfectly, by how much would the \$840 billion stimulus package have shifted the aggregate demand curve? **(2 points)**

Problem 5 (11 points)

Read the article handed out in class entitled “Think Consumption is the Engine of Our Economy? Think Again.” Write a short **summary** of the main idea of the article. This is a practice for the memo assignment that you will write later in the semester. By writing this summary now, you’ll have the opportunity to get feedback on your ability to summarize macroeconomic principles before completing the memo assignment which is worth more points.

What is a summary? A summary is a restatement in your own words. It is *not* about your feelings or your opinion. It is not a reflection piece. A summary answers the question *what is the article about?*; it does not answer the question *what do you think about this article?* nor *what is your opinion of this article?* Imagine that a friend in the college program saw you reading this article and asked you what the article was about. Your answer to your friend’s question is what you should write for this assignment.

What audience is my summary for? Your summary should be written for an audience (such as your fellow college students in the program) who are smart, but *not* necessarily familiar with macroeconomics. Therefore, your summary should introduce, explain and correctly summarize relevant economic concepts and principles that would not be necessarily be understood by someone outside of this class.

What should my summary say? Your summary should re-state the main point of the article, as well as some sub-points of the article. The article has a lot going on in it. It’s your job to figure out what the main point(s) in the article are, and condense them down for your assignment. You can’t summarize everything – so prioritize what you think is important. Since your audience is not expert, you will also want to offer definitions and explanations of key terms and concepts that are relevant to those main points. Ask yourself: If you were summarizing the article to a smart friend, where might they stop you and ask for more explanation? Where might they be confused? Those are places to offer more definitions of terms and more explanations of the economic logic. Some things in the article might not seem like things we have covered yet, which is true. But we’ve covered a lot, so there’s plenty to summarize just focusing on what you have learned in class.

Why are we doing this assignment? The purpose of this assignment is for you to practice communicating about macroeconomics using the knowledge that you’ve gained from class in written form. Answering homework questions demonstrates your knowledge, but it isn’t authentic

in terms of how you will likely apply your economic knowledge once you leave the classroom. Instead, you'll likely read articles about the economy, talk about the economy, and write about the economy. In order to summarize the article, you must understand what the article says, which means applying what you've learned in class.

Additional guidelines:

- The page limit is 3 pages. It does not need to be this long, but this is the maximum.
- Please double-space your assignment.
- Do **not** use quotations from the article. This writing assignment is sufficiently short that you should use entirely your own words.
- You do not need to provide a works cited or any citations for this assignment.
- You do not need a title or heading; just make sure that your name is on the summary.
- As noted above, this assignment is all about summary (explaining in your own words the main points of the article). This is not a writing assignment in which you are making an argument. Therefore, you do not need, and in fact *should not have*, a thesis statement.
- Furthermore, this is *not* an opinion or reflection piece. Therefore, your summary should not include your opinions about or reflections on the article.
- Your summary will be graded on the accuracy with which you capture the main points of the article, your explanation and appropriate use of relevant economic concepts, the clarity and correct use of writing conventions (grammar, spelling, etc.), and your adherence to the above guidelines.

Correcting Your Problem Set (8 points)

You'll receive 8 points for correcting your problem set in class — that is, marking the answers that you got correct on this sheet, and writing an explanation for why some of your answers were incorrect on others.